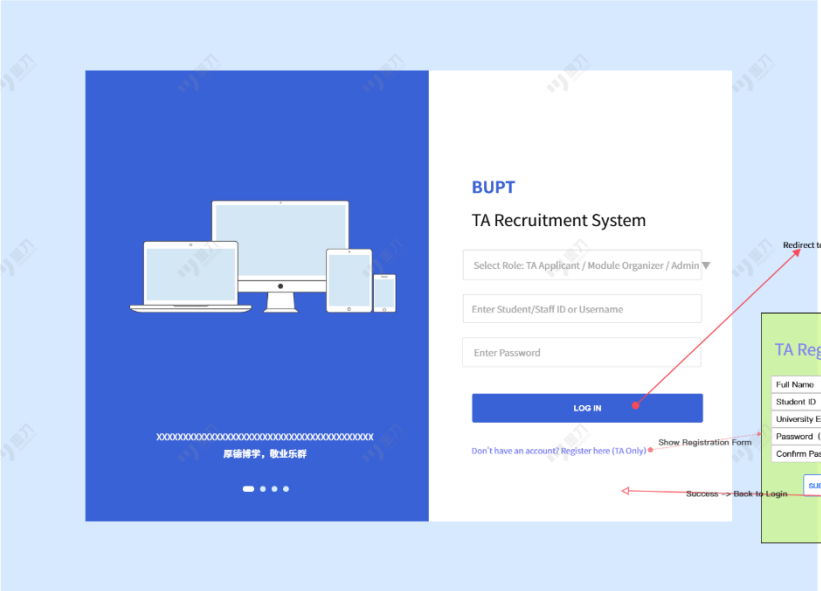
TA Recruitment System Prototype

An end-to-end system for streamlining the TA recruitment, application and management processes.

Prototype Page	Core Page Function	Corresponding Story ID	Requirement Mapping
1: Login & Registration	Role-based authentication, TA account registration	1,2,5,9	1. TA account registration (unique ID/email validation, password complexity check) 2. TA/MO/Admin login credential validation 3. Role-based system entry shunting 4. In-memory session management and logout function
2: TA Job List	TA position browsing, sorting, filtering, application initiation	3	1. Display full position info (title/module/workload/deadline/vacancy) 2. Sort positions by deadline/module name 3. Differentiate available/closed/no-vacancy position states 4. Graceful empty state handling 5. Direct application initiation for valid positions
3: TA Application Form	TA application submission, CV upload	4	1. Collect applicant info (skills, availability, work experience) 2. CV upload with format/size limit (PDF/DOCX, ≤5MB) 3. Auto-rename uploaded CV files (StudentID_Timestamp.ext) 4. Submission confirmation feedback 5. Pre-submission vacancy and deadline check
4: MO Dashboard	MO position publishing, application management, recruitment overview	6,7,8	1. TA position creation (info capture, unique ID generation) 2. View only own position applications with full applicant info 3. Application status filtering (pending/hired/rejected) 4. Application approval/rejection with status timestamp 5. Trigger notification for approved applications 6. Color-coded application status display
5: Admin User Management	Centralized user/permission management, password reset	10,14,21	1. Centralized MO/TA/Admin account management (edit/activate/suspend) 2. Role permission allocation and revocation 3. One-click password reset with automatic email notification 4. Global RBAC permission control for all operations 5. Block and log unauthorized operations 6. Prevent deletion of MO with active positions

Page 1: Login & Registration



Corresponding Story IDs: 1, 2, 5, 9
Applicable Roles: TA, MO, Admin
Core Objective: Role-based identity authentication and TA account registration, fulfilling the core acceptance criteria of account & login-related stories.

1. Overall Layout

A centralized login card (1/3 of the page) with the top title *TA Recruitment System* is designed, focusing user operations on authentication and registration, matching basic interaction requirements of login stories.

2. Role Selector

A dedicated selector (*Select Role: TA Applicant / MO / Admin*) is set at the card top, realizing role-based login shunting, and laying the foundation for role-specific dashboard loading and permission isolation (Story 2,5,9).

3. Login Module

Arranged with *Username Input* (Student/Staff ID) and *Password Input* for credential validation (Story 2,5,9). The highlighted **LOG IN** button is annotated with *Redirect to Dashboard (Based on Role)*, fulfilling role-based menu loading and session maintenance requirements.

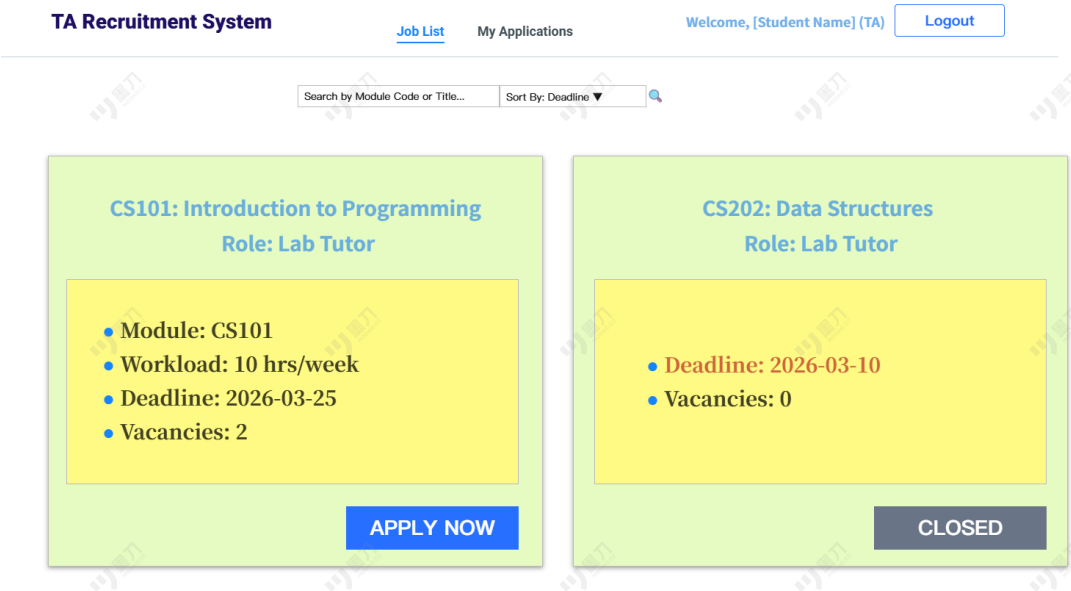
4. TA Registration Module (Story 1)

- A clickable *Register here (TA Only)* link triggers the dashed-box registration form (right/below login card) via a dashed arrow.
- The form includes 5 fields (*Full Name/Student ID/University Email/Password/Confirm Password*) for basic info capture, meeting unique ID/email validation needs.
- A password hint (*Uppercase + lowercase + numbers*) is added to meet password complexity criteria; the **SUBMIT REGISTRATION** button is annotated with *Success -> Back to Login*, forming a closed registration-login loop and fulfilling success feedback requirements.

5. Story Alignment

This page covers core criteria of Story 1 (unique validation, password complexity, feedback) and Story 2,5,9 (role-based auth, dashboard redirection, session management), and ensures isolation of different role account systems.

Page 2: TA Job List



Corresponding Story ID: 3

Applicable Role: TA Applicant

Core Objective: Realize TA position browsing, sorting, filtering and application initiation, fully meeting the acceptance criteria of position viewing requirements.

1. Global Navigation Bar

A full-width top navigation bar with system name *TA Recruitment System* (left), user info *Welcome, [Student Name] (TA)* + **Logout** button (right), and two menu links (*Job List* highlighted, *My Applications*) in the middle. It supports in-memory session maintenance and role-based identity display, and provides quick page switching.

2. Filter & Search Section

A toolbar below the navigation bar includes a search box (*Search by Module Code or Title...*), a sorting dropdown (*Sort By: Deadline* ◤) and a **Search** button. It fulfills the *sort by deadline/module name* requirement, and the dynamic filter prompt supports real-time list update for efficient position screening.

3. Job Cards List (Main Section)

2-3 vertical position cards are designed with differentiated states for available/closed positions:

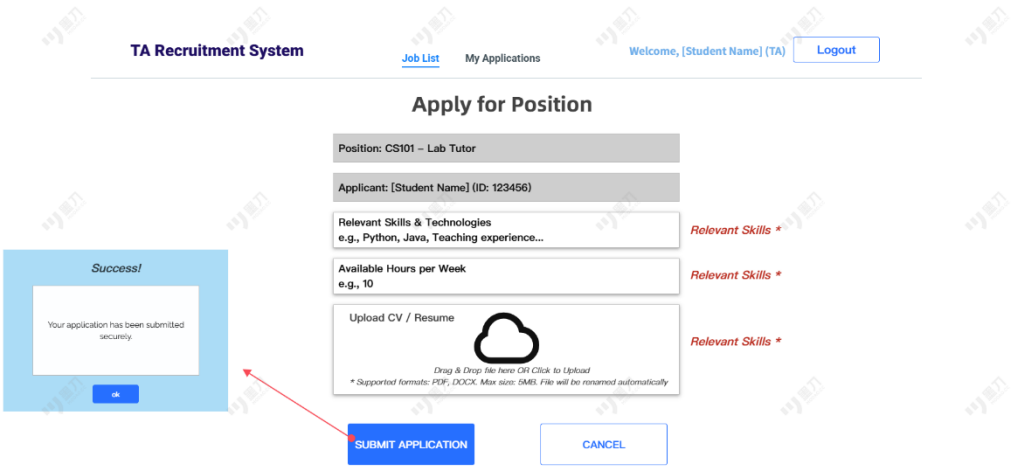
- **Available card**: Displays full position info (title/module/workload/deadline/vacancy), vacancy number marked in green, with a solid **APPLY NOW** button at the bottom right (arrow annotated *Go to Application Form (Page 3)*).
- **Closed card**: Deadline marked in red, vacancy *0* in gray/red, with a gray/dashed non-clickable button (*CLOSED/NO VACANCY*).

All cards fully display the required position details and realize **graceful empty state & status differentiation** for different vacancy/deadline conditions.

4. Story Alignment

Covers all core acceptance criteria of Story 3: complete position info display, deadline/module sorting, dynamic filter, differentiated status handling for available/closed positions, and direct application initiation via the **APPLY NOW** button, with clear interaction flow for subsequent operations.

Page 3: TA Application Form



Corresponding Story ID: 4Applicable Role: TA ApplicantCore Objective: Enable TA applicants to fill in application details and upload CVs, meeting all core acceptance criteria for application submission.

1. Global Navigation Bar

Consistent with Page 2, with system name, personalized user info, Logout button and two menu links, maintaining unified interface style and session management.

2. Form Header

Features the title Apply for Position, with two read-only fields below showing the target position and applicant ID/name (non-editable), clarifying the application object and ensuring information accuracy.

3. Core Input Fields

Three required fields (marked with red *) for application information collection:

Relevant Skills & Technologies: Multi-line text box for skill description

Available Hours per Week: Short input box for workload declaration

Previous Experience Summary: Multi-line text box for experience description

All fields capture the personal info, skills and availability required for application submission.

4. CV Upload Area (Key Module)

A dashed drag-and-drop box with Drag & Drop file here OR Click to Upload prompt, plus clear format/size hints: Supported formats: PDF, DOCX. Max size: 5MB. A side annotation notes the automatic renaming rule ([StudentID]_[Timestamp].ext), aligning with the CV storage and naming requirements of Story 4, with an error prompt (File too large or wrong format) for invalid uploads.

5. Bottom Action Buttons

Two differentiated buttons: solid/bold SUBMIT APPLICATION (primary) and hollow CANCEL (secondary), meeting basic form operation needs.

6. Interaction & Feedback

A solid arrow from SUBMIT APPLICATION triggers a success pop-up (Success! Your application has been submitted securely.) with an OK button

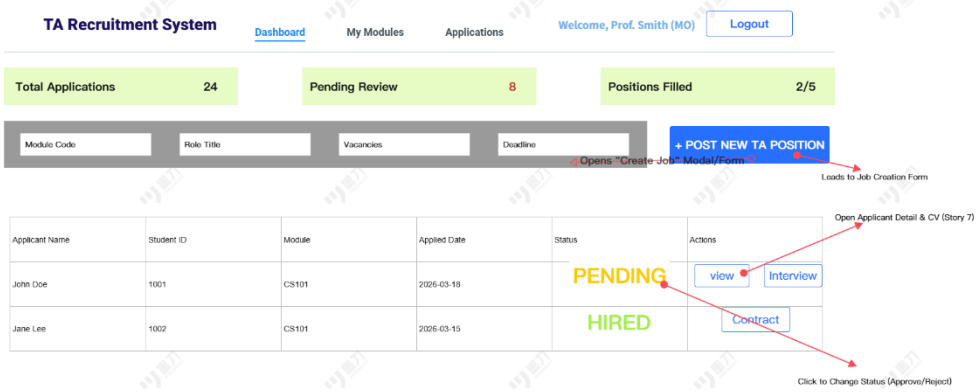
Clicking OK redirects back to Page 2 (Job List), forming a complete application-submission-feedback loop

Error prompts for invalid uploads support real-time validation, ensuring data submission compliance.

7. Story Alignment

Fully covers Story 4's acceptance criteria: complete application info capture, strict CV format/size validation, automatic file renaming, secure submission with confirmation feedback, and the atomic operation foundation for vacancy check and data storage before submission.

Page 4: MO Dashboard



Corresponding Story IDs: 6, 7
Applicable Role: Module Organizer (MO)
Core Objective: Realize MO's core operations including position publishing, application management and data overview, meeting the acceptance criteria of position creation and application viewing requirements.

1. Global Navigation Bar

MO-specific full-width navigation bar with system name (left), MO user info *Welcome, Prof. Smith (MO)* + **Logout** button (right), and three menu links (*Dashboard* highlighted). Differentiated from TA's navigation, it aligns with role-based menu loading and permission isolation requirements, maintaining unified session management.

2. Stats Overview Cards

Three horizontal small cards below the navigation bar display core data: *Total Applications*, *Pending Review* (orange number for prompt), *Positions Filled* (ratio display). Provides real-time recruitment progress overview for MOs, laying a data foundation for application review and position adjustment.

3. Post New Job Section

A prominent bold-bordered **+ POST NEW TA POSITION** button is designed, with an annotation *Opens "Create Job" Modal/Form*. The optional mini form sketch (Module Code/Role Title/Vacancies/Deadline) previews core fields of position creation, directly responding to Story 6's position information capture and unique ID generation requirements, with clear trigger logic for the creation function.

4. Applications Table (Main Section)

A structured table with headers *Applicant Name/Student ID/Module/Applied Date/Status/Actions*:

- **Status column**: Color-coded tags (*PENDING* in orange/yellow, *HIRED* in green,

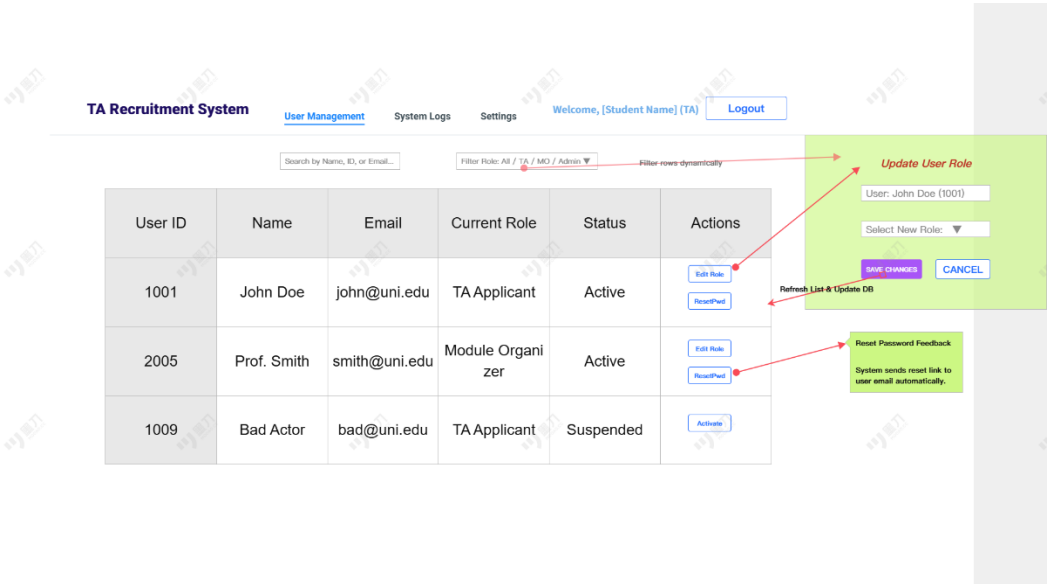
REJECTED in red) for intuitive application status display, meeting Story 7's status filtering needs; clickable PENDING tags support status modification (Approve/Reject).

- **Actions column**: Role-based operable buttons (*[View]/[Interview]* for pending applications, *[Contract]* for hired ones); the [View] button links to applicant details and CV, fulfilling Story 7's requirement of viewing own positions' application information.
- **Empty state handling**: Displays *No applications received yet* when the table is empty, aligning with graceful empty state design rules.

5. Interaction & Story Alignment

- The position publishing button triggers the creation form, covering Story 6's core criteria: required field capture, preview before publishing, unique position ID generation.
- The application table supports MO's exclusive viewing of own position applications, with status filtering, CV viewing and status modification functions, fully meeting Story 7's acceptance criteria of permission filtering and application information display.
- Color-coded status and operable action buttons realize the basic status tracking and operation logic of MO's application management, laying the foundation for subsequent approval/rejection operations.

Page 5: Admin User Management



Corresponding Story ID: 10

Applicable Role: Admin

Core Objective: Realize admin's centralized management of all user accounts, role adjustment and password reset, meeting the acceptance criteria of MO account management and permission control requirements.

1. Global Navigation Bar

Admin-exclusive full-width navigation bar with system name (left), admin identity info *Welcome, Admin (System)* + **Logout** button (right), and three menu links (*User Management* highlighted). Differentiated from TA/MO navigation, it matches the highest access level design for admin, supporting role-based permission isolation and session maintenance.

2. Search & Filter Section

A toolbar below the navigation bar includes a search box (*Search by Name, ID, or Email...*) and a role filter dropdown (*Filter Role: All / TA / MO / Admin* ▼). The dynamic filter function realizes real-time list screening by role, improving the efficiency of admin's batch user management, with clear interaction annotation for dynamic list update.

3. User List Table (Main Section)

A structured table with headers *User ID/Name/Email/Current Role/Status/Actions*, the core carrier of centralized user management:

- **Role/Status display**: Clearly shows user's current role and account status, with color coding (*Active* in green, *Suspended* in red) for intuitive identification; status differentiation matches the requirement of preventing deletion of MO with active positions.
- **Action buttons**: Role/status-based operable buttons (*[Edit Role] [ResetPwd]* for active users, *[Activate]* for suspended users), realizing differentiated management operations and meeting the create/edit/delete requirements of user account management.
- **Data rows**: Typical examples of TA/MO/suspended users are designed, covering the main user types managed by admin, with consistent layout and clear information hierarchy.

4. Key Functional Modules

4.1 Edit Role Modal

A pop-up modal (dashed/superimposed box) triggered by *[Edit Role]* button, titled *Update User Role*, displays the target user info (*User: John Doe (1001)*) and a new role selector (*Select New Role: TA, MO, Admin, None*), with **SAVE CHANGES** (solid) and **CANCEL** (hollow) buttons. The modal realizes role adjustment and permission allocation/revocation, and the *Refresh List & Update DB* annotation ensures data consistency after modification, aligning with the JSON data update and write lock requirements.

4.2 Reset Password Function

An arrow from the *[ResetPwd]* button points to a prompt bubble (*System sends reset link to user email automatically*), realizing one-click password reset with automatic notification. The design matches the password reset logic and operation logging requirements, and the automated process reduces admin's manual operation costs.

5. Interaction & Story Alignment

- Fully covers Story 10's core acceptance criteria: centralized create/edit/delete of user (MO) accounts, role permission adjustment, differentiated management of account status, and data consistency after operation (refresh list & update DB).
- The role filter, search function and differentiated action buttons improve the efficiency of admin's user management, matching the requirement of admin's highest access level and comprehensive permission control.
- The pop-up modal for role editing and automatic feedback for password reset realize

closed-loop management operations, with clear interaction logic laying the foundation for operation logging and data write lock implementation.